

TCS349: Responsible and Explainable AI

Unit 3, Lecture 6

AI in Governance: Public Services & Algorithmic Administration

How governments use AI to deliver services, support policy decisions and administer programmes at scale - and the distinct accountability challenges this raises.

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Government Adoption



Policy Decision Support



**Algorithmic
Administration**



Policy Case Studies

1. AI in Government - Why It's Being Adopted

Governments face growing demand for services with limited staff and budget, making AI attractive for handling scale efficiently.

AI can help process large volumes of applications, claims or requests faster and more consistently than fully manual review.

Data-driven decision support can help policymakers identify patterns and predict outcomes across large, complex public systems.

Adoption is uneven across governments, often depending on technical capacity, funding and public trust in the institution.

2. AI in Public Services - Common Applications

Benefits and welfare administration: AI systems help determine eligibility and detect fraud in large-scale social programmes.

Public health: AI models support disease surveillance, resource allocation and outbreak prediction.

Transportation and infrastructure: AI optimises traffic signal timing, public transit scheduling and infrastructure maintenance planning.

Citizen services: chatbots and automated systems handle routine enquiries, freeing staff for more complex cases.

3. AI for Policy Decision Support

Decision-support systems analyse large datasets to help policymakers understand the likely outcomes of different policy options.

These tools can simulate the effects of a proposed policy change before it is implemented at full scale.

Decision support is meant to inform human policymakers, not replace their judgement or democratic accountability.

The quality of policy decision support depends heavily on the quality and representativeness of the underlying data, echoing Unit I's bias concerns.

4. Algorithmic Administration - Automating Government Decisions

Algorithmic administration refers to using AI systems to directly make or heavily influence individual administrative decisions, such as benefit approvals.

These systems can increase consistency and processing speed compared to fully manual case-by-case review.

Because these decisions affect legal rights and entitlements, they carry a higher accountability bar than many private-sector uses of AI.

Citizens affected by an algorithmic decision generally have a right to understand and challenge it, directly invoking Unit II's transparency and accountability principles.

5. Policy Case Study: Algorithmic Benefits Determination

Several welfare agencies have used AI systems to flag applications as high-risk for fraud, triggering additional review or suspension of benefits.

In some documented cases, flawed risk models wrongly flagged thousands of legitimate claimants, causing significant hardship before the errors were corrected.

These cases often stemmed from training data that poorly represented the true diversity of legitimate claimants.

The episodes strengthened calls for mandatory human review and a clear appeals process before any benefit is reduced or denied.

6. Policy Case Study: Predictive Policing and Risk Assessment

Some police departments have used predictive models to forecast crime hotspots or assess a defendant's risk of reoffending.

Critics have shown that these tools can reproduce and reinforce historical policing bias present in the underlying arrest data.

Legal challenges in some jurisdictions have required greater transparency about how these risk scores are calculated and used in sentencing.

This case connects directly to the bias-detection techniques from Unit I and the surveillance concerns from Unit II.

7. Governance Challenges Specific to Public-Sector AI

Due process: government decisions must be explainable and contestable in ways that go beyond typical private-sector customer service.

Democratic accountability: elected officials, not just technical teams, are ultimately answerable for algorithmic administrative decisions.

Equity of access: public AI systems must serve populations with uneven digital literacy and internet access.

Procurement and vendor oversight: many governments rely on external vendors for AI systems, raising questions about transparency into proprietary models.